UTD HACKEDGE 2026 • EDGE AI TRACK

BreathBreak

An edge-powered wearable that detects worker fatigue in real time



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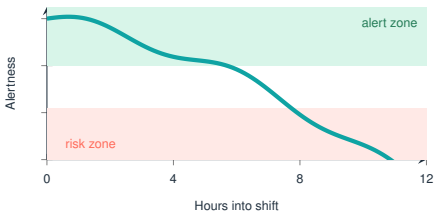
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Problem and motivation

The problem

Employees working **long shifts** often experience fatigue, which reduces alertness, harms their health, and raises the risk of **workplace accidents**.

Illustrative: alertness drifts down over a long shift



Why it matters

Productivity

Timely breathing breaks help workers regain focus, maintain performance, and work more efficiently.

Safety

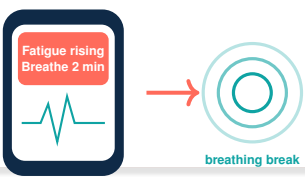
Early fatigue detection helps prevent workplace injuries and reduces costly accidents.

BreathBreak: real-time fatigue detection at the edge

A **non-invasive wearable** that monitors workers and detects signs of fatigue during long shifts.

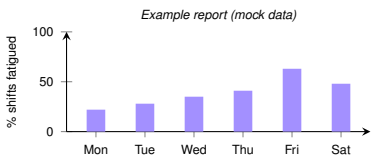
① Real-time fatigue alerts

Detects rising fatigue and prompts the worker to take a short breathing break.



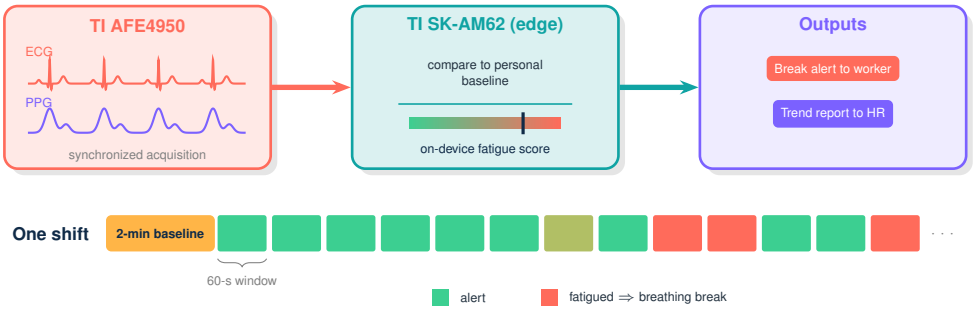
② Workload reports for HR

Weekly and monthly fatigue trends help HR balance workload and shift schedules.



Goal: higher productivity • safer workplaces • healthier employees

Technology behind BreathBreak



Task: classify each 60-s window as *alert* or *fatigued* by comparing it with the person's own 2-min baseline; alert only on *serious, sustained* fatigue.

Data: **our own AFE4950 recordings** (Model A), plus public **WESAD** (stress) and **DROZY** (sleep deprivation) for Model B.

Fatigue models

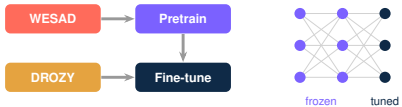
Model A — our recordings



Many people × many sessions, recorded on BreathBreak's own sensor

Pro: same sensor and same task as BreathBreak.
Con: small dataset (we record it ourselves).

Model B — transfer learning



Pro: much more data to borrow from.
Con: stress ≠ fatigue, and public sensors differ from ours.



Scalability

BreathBreak **doesn't change regular hours**. It offers short breaks **only when body signals show they're needed**.

Productivity and safety matter in long-shift work across industries:



Nurses



Truck drivers



Factory workers



& more

"A small pause is better than a big accident."